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# Problem 1: Analysis of Stock Returns

## A. Arithmetic Returns

Last five rows:

| Date       | SPY       | AAPL      | EQIX      |
|------------|-----------|-----------|-----------|
| 2024-12-27 | -0.011492 | -0.014678 | -0.006966 |
| 2024-12-30 | -0.012377 | -0.014699 | -0.008064 |
| 2024-12-31 | -0.004603 | -0.008493 | 0.006512  |
| 2025-01-02 | -0.003422 | -0.027671 | 0.000497  |
| 2025-01-03 | 0.011538  | -0.003445 | 0.015745  |

Standard deviations:

- SPY: 0.008077
- AAPL: 0.013483
- EQIX: 0.015361

## B. Log Returns

Last five rows:

| Date       | SPY       | AAPL      | EQIX      |
|------------|-----------|-----------|-----------|
| 2024-12-27 | -0.011515 | -0.014675 | -0.006867 |
| 2024-12-30 | -0.012410 | -0.014696 | -0.007972 |

|            |               |               |          |
|------------|---------------|---------------|----------|
| 2024-12-31 | -0.00457<br>7 | -0.00842<br>7 | 0.006602 |
| 2025-01-02 | -0.00339<br>2 | -0.02793<br>0 | 0.000613 |
| 2025-01-03 | 0.011494      | -0.00335<br>6 | 0.015725 |

#### Standard deviations:

- SPY: 0.008078
- AAPL: 0.013446
- EQIX: 0.015270

## Problem 2: Portfolio Value-at-Risk Analysis

### A. Portfolio Value Calculation (as of 2025-01-03)

Total portfolio value: \$251,862.50

### B. VaR and ES Calculations (5% alpha level, arithmetic returns, 0 mean)

#### Method A: Normal Distribution with Exponentially Weighted Covariance ( $\lambda=0.97$ )

| Asset     | VaR (\$)     | ES (\$)      |
|-----------|--------------|--------------|
| SPY       | 827.85       | 1,038.1<br>6 |
| AAPL      | 946.08       | 1,186.4<br>2 |
| EQIX      | 2,933.5<br>1 | 3,678.7<br>4 |
| Portfolio | 3,856.3<br>2 | 4,835.9<br>8 |

### Method B: T-Distribution with Gaussian Copula

| Asset     | VaR (\$) | ES (\$)  |
|-----------|----------|----------|
| SPY       | 765.86   | 1,025.47 |
| AAPL      | 1,032.33 | 1,459.92 |
| EQIX      | 3,436.30 | 4,884.37 |
| Portfolio | 4,379.51 | 6,123.02 |

### Method C: Historical Simulation (Full History)

| Asset     | VaR (\$) | ES (\$)  |
|-----------|----------|----------|
| SPY       | 872.40   | 1,080.10 |
| AAPL      | 1,067.11 | 1,437.79 |
| EQIX      | 3,635.08 | 4,714.89 |
| Portfolio | 4,575.03 | 6,059.39 |

## C. Discussion of Differences Between Methods

1. Method A (Normal Distribution) has the lowest portfolio VaR and ES estimates. This approach assumes returns follow a normal distribution with exponentially weighted covariance. It gives more weight to recent observations. It is easy to calculate but often underestimates tail risk because financial returns usually have fatter tails than the normal distribution.
2. Method B (T-Distribution with Gaussian Copula) gives VaR and ES values that are in between the other two methods. This method captures the heavy tails of asset returns better by using the t-distribution. It uses a Gaussian copula to model how assets depend on each other. The higher ES values compared to Method A show that it does a better job of accounting for large losses.
3. Method C (Historical Simulation) gives the highest VaR estimates but slightly lower ES than Method B. This method does not assume any distribution. Instead, it uses historical returns directly. It shows real market behaviors, including extreme events from the historical data. However, it depends heavily on the historical period chosen.

# Problem 3: Option Pricing and Risk Analysis

## A. Implied Volatility

Implied Volatility = 33.51%

**B. Calculate Delta, Vega, and Theta. Using this information, by approximately how much would the price of the option change if the implied volatility increased by 1%. Prove it.**

- Delta = 0.6659
- Vega = 5.6407
- Theta (annual) = -5.5446

Approximate price change if volatility increases by 1% = \$0.0564

First, I calculated the call option price using the original implied volatility of 33.51%, which gave us \$3.00 (the given call price)

Then, I calculated the call option price using an increased implied volatility of 34.51% (original + 1%), which resulted in a price of \$3.0564

The difference between these two prices (\$3.0564 - \$3.00) is \$0.0564, which represents the actual price change due to a 1% increase in volatility

**C. Calculate the price of the put using Generalized Black Scholes Merton. Does Put-Call Parity Hold?**

- Put Price = \$1.2593
- Put-Call Parity difference = 0.0000e+00

Yes, Put-Call Parity holds perfectly. When we verify the put-call parity relationship using the formula:  $c + Ke^{-rT} = p + S$ . And substitute our values:  $\$3.00 + \$30e^{(-0.10 \times 0.25)} = \$1.2593 + \$31$   $\$3.00 + \$29.2593 = \$1.2593 + \$31$   $\$32.2593 = \$32.2593$  The difference between the left and right sides is exactly 0.0000e+00, confirming that put-call parity holds with no discrepancies.

## D. Portfolio Risk Analysis

### Portfolio Composition (today):

- Call Value = \$3.0000
- Put Value = \$1.2593
- Stock = \$31.0000
- Total Portfolio Value = \$35.2593

### Portfolio Greeks:

- Portfolio Delta = 1.3319
- Portfolio Theta = -8.1632

### Risk Measures (20-day, 5% level):

#### Delta-Normal Approximation:

- Mean change = -\$0.6403
- Standard deviation of change = \$2.8907
- VaR = \$5.3951
- ES (Standard) = \$6.6030

#### Monte Carlo Simulation:

- VaR = \$4.1886
- ES = \$4.6216

## E. Comparison of Methods

1. The blue line shows the actual portfolio value. Options do not change in value evenly with stock prices.
2. Delta-Normal Approximation Limitations: The orange dashed line is the Delta-Normal method. This method uses a straight line based on a delta of 1.3319 to estimate portfolio value. This method:
  - Works well near the current stock price (\$31)
  - Greatly underestimates portfolio value at higher stock prices (above \$37)
  - Also underestimates portfolio value at lower stock prices (below \$27)
3. Monte Carlo Simulation Advantages: The green dots represent the Monte Carlo simulation results. This method:
  - Matches the actual portfolio values closely for all stock prices
  - Captures the curved shape of the relationship between the portfolio and stock prices
  - Takes into account the effect of time decay in options over 20 days
  - Correctly includes the range of possible outcomes

### Differences in Risk Measures:

- Delta-Normal shows higher VaR (\$5.40 vs. \$4.19) and ES (\$6.60 vs. \$4.62)
- Delta-Normal method's straight-line approach overestimates the risk of large losses because the line becomes inaccurate when far from the current price
- Monte Carlo provides better risk estimates because it fully recalculates option values at each simulated stock price

